

## Summary

Productivity and efficiency analysis can be used to guide policy development by providing insight into the relative performance of firms, and the drivers of industry growth and competitiveness. In this report, ABARES has applied stochastic frontier analysis to a novel dataset of sawmilling firms in 2016–17, allowing productivity to be measured and decomposed into various measures of efficiency. This study is the first comprehensive productivity and efficiency analysis of the Australian sawmilling industry at the mill level.

Productivity measurement is used to assess the physical performance of mills and is primarily concerned with the quantities of outputs produced and inputs used by firms, and is largely independent of prices. This concept should not be confused with profitability, which is a measure of financial performance, dependent on both quantities and prices.

This report includes estimates of productivity, technical efficiency, and scale and mix efficiency. Measures of productivity indicate how well mills have converted inputs into outputs. Technical efficiency is a measure of how well production technologies are chosen and used by mills. Scale and mix efficiency is a measure of how well mills have captured economies of scale or substitution (that is, changed the size of the mill, or changed the mix of inputs used or outputs produced to maximise productivity). Analysing the individual components of efficiency provides greater insight into firm performance than considering productivity estimates in isolation.

On average, hardwood mills were 48% as productive as the most productive hardwood mill in 2016–17 and softwood mills were 75% as productive as the most productive softwood mill. Variation in productivity between hardwood mills was substantially greater than the variation observed in the softwood sector.

Softwood mills were estimated to have higher average technical efficiency than hardwood mills in 2016–17 (86% compared to 78%). The difference between the most and least technically efficient hardwood mills was much greater than the difference in the softwood sector. A small number of relatively low-productivity mills were a significant reason for the lower average technical efficiency of the hardwood sector.

On average, hardwood mills were 60% scale and mix efficient (compared to the most scale and mix efficient hardwood mill) in 2016–17 and softwood mills were 71% scale and mix efficient (compared to the most scale and mix efficient softwood mill). There was a larger gap between the most and least scale and mix efficient hardwood mills than for the softwood mills.

Productivity tended to increase in line with mill size (volume of sawlogs processed) and was largely driven by scale and mix efficiency. Technical efficiency did not always increase as mill size increased—relatively high levels of technical efficiency were observed in mills of all sizes and many smaller mills were more technically efficient than Australia's largest mills.

Recent policy measures, such as the National Forest Industries Plan, have focused on expanding production forest area to promote industry growth. This report indicates that efforts to increase sawmill productivity and efficiency also have significant potential to increase output and improve international competitiveness. The positive relationship between mill size and productivity in this study suggests that mill consolidation could also improve these outcomes.

## **Read the full report**

[Productivity and efficiency of the Australian sawmilling industry – PDF \[1.5 MB\]](#)

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